

Infrared Electromagnetic Criticality and a Conformal Crossover

Experiments, analytic findings, and formalisation targets

Working note prepared from the preceding investigation

14 September 2026

Abstract

We study a minimal linear electromagnetic-response model in an asymptotically de Sitter background, its nonlinear gauge-invariant completion, and a reciprocal conformal matching between consecutive aeons. The linear characteristic polynomial is obtained exactly. Under positive parameters, loss of stability through a zero eigenvalue occurs first at zero physical wavenumber, and the unstable band is infrared. A scalar-dependent Maxwell kinetic function can subsequently amplify finite-wavenumber photons. Four-dimensional Maxwell theory transports regular conformal field data across a conformal boundary, while reciprocal scale factors map an outgoing de Sitter asymptotic law to an incoming radiation-era scale-factor law. We distinguish algebraic consequences from dynamical assumptions: the construction does not yet derive the nonlinear saturation amplitude, the conformal-factor normalization, entropy production, or a complete semiclassical crossover.

1 Status of the claims

The calculations support a conditional mechanism, not a completed cosmological theory. It is useful to separate three levels:

Status	Statement
Derived	The characteristic polynomial, the stationary threshold, monotonic decrease of the stationary gain with q^2 , the infrared unstable band, and reciprocal scale-factor asymptotics.
Model-dependent	A scalar response h with $I(h)^2 F^2$ coupling can pump electromagnetic modes; oscillations can yield Hill/Mathieu resonance.
Open	Whether the instability reaches the required conformal radiation amplitude, whether backreaction thermalises, and whether a regular solution of the gravitational matching problem fixes the reciprocal normalization.

2 Linear electromagnetic experiment

For one helical Fourier mode with physical wavenumber $q = k/a$ and helicity $s \in \{-1, +1\}$, consider

$$\dot{E} = -2HE + sqB + \alpha h, \quad (1)$$

$$\dot{B} = -2HB - sqE, \quad (2)$$

$$\dot{h} = \sigma E - \gamma h. \quad (3)$$

Here $H > 0$ is treated as constant, $\gamma > 0$ is the response relaxation rate, and only the product $A := \alpha\sigma$ enters the spectrum. The Jacobian is

$$J = \begin{pmatrix} -2H & sq & \alpha \\ -sq & -2H & 0 \\ \sigma & 0 & -\gamma \end{pmatrix}.$$

Direct expansion of $\det(\lambda I - J)$ gives

$$p_q(\lambda) = (\lambda + \gamma)((\lambda + 2H)^2 + q^2) - A(\lambda + 2H). \quad (4)$$

The helicity cancels, so both helicities have the same spectrum.

At a stationary crossing, $\lambda = 0$. Equation (4) yields

$$A = \gamma \left(2H + \frac{q^2}{2H} \right). \quad (5)$$

Define the stationary gain

$$G_{\text{EM}}(q) = \frac{A}{\gamma(2H + q^2/(2H))}.$$

For $A, H, \gamma > 0$,

$$\frac{\partial G_{\text{EM}}}{\partial(q^2)} = -\frac{A}{2H\gamma(2H + q^2/(2H))^2} < 0.$$

Thus the stationary threshold is reached first at $q = 0$. Equivalently,

$$\gamma_c(q) = \frac{2HA}{4H^2 + q^2}, \quad \gamma_c(q) \leq \gamma_c(0) = \frac{A}{2H}.$$

When $A > 2H\gamma$, the value $p_q(0)$ is negative precisely for

$$0 \leq q^2 < q_c^2, \quad q_c^2 = \frac{2HA}{\gamma} - 4H^2. \quad (6)$$

Since $p_q(\lambda) \rightarrow +\infty$ as $\lambda \rightarrow +\infty$, $p_q(0) < 0$ guarantees a positive real root. Hence (6) is a rigorous sufficient instability test and, for a zero-mode stationary crossing, gives the exact infrared band.

Qualification. The zero-eigenvalue test alone does not exclude an earlier Hopf crossing for arbitrary parameters. Applying the cubic Routh–Hurwitz criterion to (4) is required for a global “first instability” theorem along any specified parameter path. The IR-first conclusion proved here concerns the stationary branch and the common experiment in which γ decreases through the zero-mode threshold.

3 Near-threshold growth

Write $A = 2H\gamma(1 + \epsilon)$ with $0 < \epsilon \ll 1$. Taylor expansion of $p_q(\lambda) = 0$ around $(\lambda, q^2, \epsilon) = (0, 0, 0)$ gives

$$\lambda(q) = \frac{4H^2\gamma\epsilon - \gamma q^2}{4H^2 + 4H\gamma - A} + O(\epsilon^2, \epsilon q^2, q^4).$$

Replacing the denominator by its critical value gives

$$\lambda(q) \simeq \frac{\gamma}{2H(2H + \gamma)}(4H^2\epsilon - q^2). \quad (7)$$

The leading unstable band is $q < 2H\sqrt{\epsilon}$, and the maximum occurs at $q = 0$. This is an infrared onset rather than an ultraviolet catastrophe.

4 Nonlinear gauge-invariant completion

A minimal covariant completion uses a scalar response and a field-dependent Maxwell kinetic function,

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2}(\nabla h)^2 - V(h) - \frac{1}{4}I(h)^2 F_{\mu\nu} F^{\mu\nu} \right]. \quad (8)$$

For example, a post-critical Landau potential

$$V(h) = -\frac{1}{2}\mu^2 h^2 + \frac{\lambda_h}{4}h^4, \quad \mu^2, \lambda_h > 0,$$

saturates at $|h_\star| = \mu/\sqrt{\lambda_h}$ in the absence of strong backreaction.

In conformal FLRW coordinates and Coulomb gauge, a transverse Fourier mode with canonical variable $u_k = IA_k$ obeys

$$u_k'' + \left(k^2 - \frac{I''}{I} \right) u_k = 0. \quad (9)$$

Consequently, a non-adiabatic homogeneous response is a pump. A sufficient interval-wise condition for tachyonic growth is $I''/I > k^2$. If h oscillates about its minimum, the periodic part of I''/I produces a Hill equation and may generate parametric resonance. This provides a concrete channel from an infrared response instability to finite- k photon occupation.

This is not a proof of reheating. Efficiency depends on the profile of $I(h)$, expansion, rescattering, strong-coupling constraints, and backreaction. The electromagnetic energy must be computed self-consistently with the h equation.

5 Conformal matching experiment

Let a regular “bridge” metric $\hat{g}$ describe a neighbourhood of the crossover Σ , and write

$$g_{\mu\nu}^- = \Omega_-^2 \hat{g}_{\mu\nu}, \quad g_{\mu\nu}^+ = \Omega_+^2 \hat{g}_{\mu\nu}.$$

Assume $\Omega_- \rightarrow \infty$, $\Omega_+ \rightarrow 0$, and impose the reciprocal ansatz

$$\Omega_+ \Omega_- = C_0, \quad C_0 > 0. \quad (10)$$

In four spacetime dimensions, source-free Maxwell equations are conformally invariant when the two-form $F_{\mu\nu}$ is held fixed. The covariant stress tensor and measured density scale as

$$T_{\mu\nu} = \Omega^{-2} \hat{T}_{\mu\nu}, \quad \rho = \Omega^{-4} \hat{\rho}.$$

Thus finite nonzero conformal data $\hat{\rho}$ appear cold on the outgoing side and hot on the incoming side:

$$\rho_- \rightarrow 0, \quad \rho_+ \rightarrow \infty.$$

This is a statement about the relation between two physical metrics and common conformal data; it must not be interpreted as ordinary energy amplification within one metric.

For a spatially flat outgoing de Sitter asymptotic,

$$a_-(\eta) = \frac{1}{H_-|\eta|} + o(|\eta|^{-1}), \quad \eta \rightarrow 0^-,$$

unit reciprocity gives, after identifying positive conformal time on the incoming side,

$$a_+(\eta) = \frac{1}{a_-(-\eta)} = H_- \eta + o(\eta).$$

Since $dt_+ = a_+ d\eta$, this implies

$$a_+(t_+) \propto t_+^{1/2},$$

the radiation-era FLRW exponent.

Normalization caveat. Reciprocity alone fixes an asymptotic form only up to normalization. If one additionally assumes a flat radiation-dominated Friedmann equation and writes $a_+ = C\eta$, then

$$\hat{\rho}_\gamma = \frac{3C^2}{8\pi G}.$$

Setting $C = H_-$ by a chosen unit reciprocity gives $\hat{\rho}_\gamma = 3H_-^2/(8\pi G)$. This equality is therefore a consistency condition involving the conformal gauge and Einstein dynamics, not an independently derived prediction of Maxwell conformal invariance.

6 What has and has not been closed

The calculation establishes a coherent conditional chain:

$$\begin{aligned} \text{IR stationary instability} &\longrightarrow h(t) \longrightarrow I(h) \text{ pump} \longrightarrow \text{photon production,} \\ &\longrightarrow \text{regular conformal Maxwell data} \longrightarrow a_+(t) \propto t^{1/2}. \end{aligned}$$

The remaining make-or-break tasks are:

1. derive the nonlinear saturation value $\hat{\rho}_{\text{EM}}(\Sigma)$ rather than prescribe it;
2. solve the coupled h -Maxwell-Einstein system including backreaction;
3. specify a regular conformal-factor equation and junction conditions;
4. demonstrate entropy production and an approximately thermal, isotropic state;
5. test strong coupling and quantum consistency of $I(h)$ near the crossover.

7 Lean formalisation scope

The accompanying Lean file formalises the algebraic core over $\mathbb{R}$:

- evaluation of the characteristic polynomial at zero;
- equivalence of the zero-mode threshold and $A = 2H\gamma$;
- monotonicity of the stationary critical relaxation rate in q^2 ;
- equivalence between the sign test $p_q(0) < 0$ and the infrared-band inequality;
- reciprocal scale-factor and density-scaling identities;
- the cosmic-time exponent obtained from $a = C\eta$.

It intentionally does not formalise spectral continuity, PDE existence, quantum particle production, or gravitational junction regularity.

Conclusion

The strongest defensible result is conditional but sharp: a minimal positive-parameter electromagnetic response has an infrared stationary instability; a scalar kinetic coupling supplies a gauge-invariant route to finite-wavenumber photons; and reciprocal conformal matching maps outgoing de Sitter scale-factor asymptotics to the radiation-era power law. The central unresolved quantity is the dynamically selected conformal radiation amplitude. Until it is derived in the fully coupled theory, the proposal is a mathematically structured mechanism rather than a completed cyclic cosmology.